ACS APPLIED MATERIALS & INTERFACES

Research Article

www.acsami.org

New Insights on Electro-Optical Response of Poly(3,4-ethylenedioxythiophene):Poly(styrenesulfonate) Film to Humidity

Eric S. Muckley,*[,†,‡] Christopher B. Jacobs,[†] Keith Vidal,[†] Jyoti P. Mahalik,[§] Rajeev Kumar,[†,§] Bobby G. Sumpter,[†,§] and Ilia N. Ivanov*[,†]

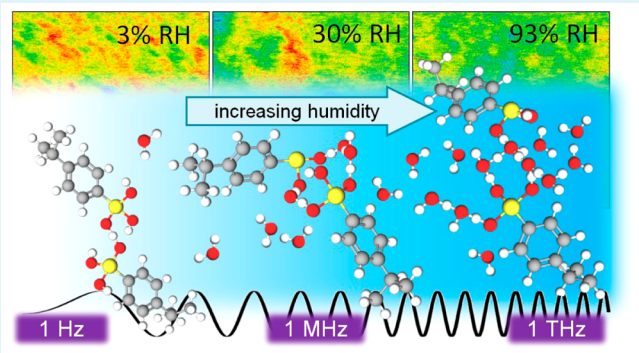
[†]Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States

[‡]Bredesen Center, University of Tennessee, Knoxville, Tennessee 37996, United States

[§]Computer Science and Mathematics Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States

Supporting Information

ABSTRACT: Understanding the relative humidity (RH) response of poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is critical for improving the stability of organic electronic devices and developing selective sensors. In this work, combined gravimetric sensing, nanoscale surface probing, and mesoscale optoelectronic characterization are used to directly compare the RH dependence of electrical and optical conductivities and unfold connections between the rate of water adsorption and changes in functional properties of PEDOT:PSS film. We report three distinct regimes where changes in electrical conductivity, optical conductivity, and optical bandgap are correlated with the mass of adsorbed water. At low (RH < 25%) and high (RH > 60%) humidity levels, dramatic changes in electrical, optical, and structural properties occur, while changes are insignificant in mid-RH (25 < RH < 60%) conditions. We associate the three regimes with water adsorption at hydrophilic moieties at low RH, diffusion and swelling throughout the film at mid-RH, and saturation of the film by water at high RH. Optical film thickness increased by 150% as RH was increased from 9 to 80%. Low frequency (1 kHz) impedance increased by ∼100%, and film capacitance increased by ∼30% as RH increased from 9 to 80% due to an increase in the film dielectric constant. Changes in electrical and optical conductivities concomitantly decrease across the full range of RH tested.



KEYWORDS: *humidity, adsorption, PEDOT:PSS, QCM, ellipsometry, polymer*

INTRODUCTION

Advances in flexible electronics facilitate the development of personal wearable sensors for environmental monitoring and biometrics.[1] The polymer mixture poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is suitable for use as a low-cost wearable relative humidity (RH) sensor because of its resistive RH response,[2] compatibility with inkjet printing,[3] and mechanical flexibility.[4,5] Its hydroscopic nature, however, limits the stability of flexible optoelectronic devices,[6,7] as exposure to water can alter electronic,[8,9] optical,[10] morphological,[11] and viscoelastic properties of PEDOT:PSS.[12−15] Despite the importance of PEDOT:PSS in flexible sensors, electrodes,[16,17] and soft actuators,[18,19] the structure of the film, the mechanism of charge carrier transport, and correlation between optical, electronic, and mechanostructural responses to RH remain poorly understood.[20] Unfolding these relationships is critical for understanding sensing mechanisms in PEDOT:PSS and improving selectivity, sensitivity, and durability of wearable RH sensors and other PEDOT:PSS-based devices.

The PEDOT:PSS polyelectrolyte complex consists of a phase segregated structure in which ∼30 nm diameter conductive PEDOT-rich polycationic domains are encapsulated by ∼1 nm thick PSS-rich polyanionic shells.[13] The domains are embedded in an electronically insulating PSS matrix loosely cross-linked by hydrogen bonding.[21] In the presence of water vapor, H_2O molecules adsorb directly onto hydrophilic PSS sulfonic acid groups: $H_2O + PSS\ (HSO_3) \rightarrow H_3O^+ + PSS(SO_3)^-$.[22] Additional water molecules hydrogen bond to less active PSS sites and previously adsorbed H_2O molecules, facilitating prompt diffusion of water into the film.[10,11] Hydrogen bonding between water molecules strains existing hydrogen bonds in the PSS matrix, which leads to interdomain decoupling and decohesion of the PEDOT:PSS complex, decreasing rigidity of the polymer matrix and swelling the film.[22] Loading of water to the point of saturation may lead to irreversible morphological changes and increases in film thickness by

Received: March 3, 2017
Accepted: April 26, 2017
Published: April 26, 2017

ACS Publications © XXXX American Chemical Society

DOI: 10.1021/acsami.7b03128
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

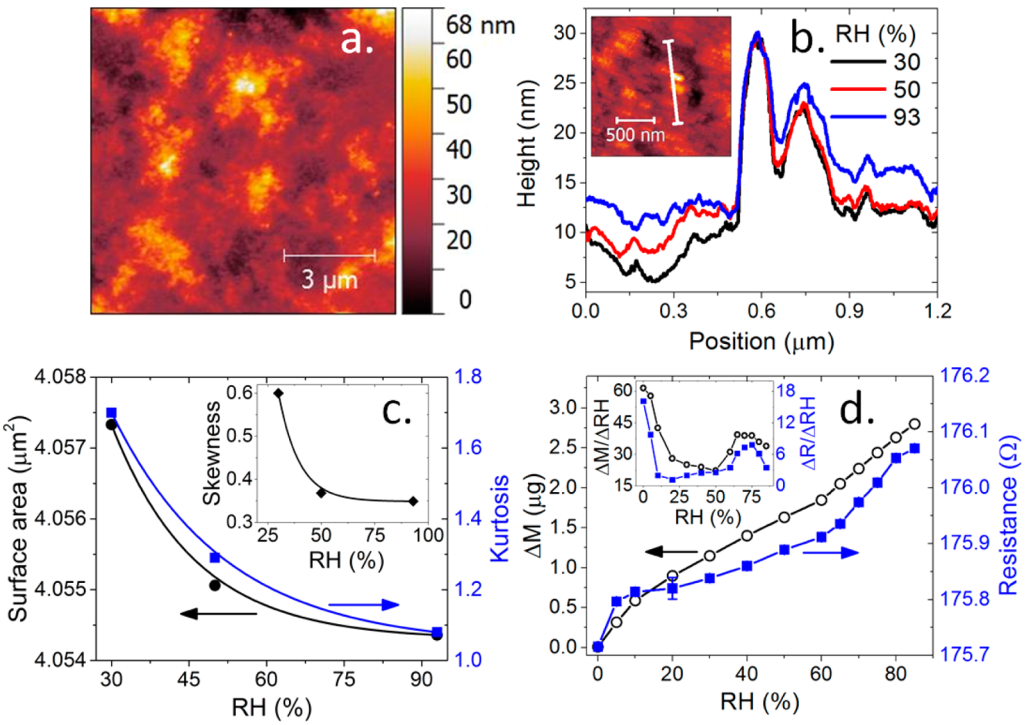


Figure 1. (a) AFM image of PEDOT:PSS film at ambient humidity. (b) Cross-section of PEDOT:PSS film at 30, 50, and 93% RH. Average height of the film surface increases relative to the height of the tallest feature with increasing RH. Inset shows the cross section region. (c) Statistical analysis of the image in panel b at 30, 50, and 93% RH. Inset: skewness in the surface height distribution. (d) Mass change (ΔM, black open circles) and electrical resistance (solid blue squares) of PEDOT:PSS at different RH levels. Inset: average rate of change in mass ($\Delta M/\Delta$RH) and average rate of change in resistance ($\Delta R/\Delta$RH) as a function of changing RH levels. The units of $\Delta M/\Delta$RH are ng/%RH, and $\Delta R/\Delta$RH units are mΩ/%RH.

75% under some conditions[10,23] while showing only 2% volume change in other experiments.[24] The degree of film swelling is dependent on experimental conditions, the ratio of PEDOT to PSS in the film, film aspect ratio, and the presence of dopants. Strong van der Waals and hydrogen bonding between water molecules becomes a dominant force after H_2O monolayer formation,[25] and exposure to RH levels above 80% leads to H_2O multilayer formation and condensation on the PEDOT:PSS surface.[2,6]

PEDOT:PSS is most often used as a hole transport layer in organic electronic devices, and recently, transport of ions injected from electrolyte solutions was studied.[26] The conductivity of PEDOT:PSS at RH < 30% is thought to be mediated by mobility of holes between conductive PEDOT-rich domains. The hydrophilic PSS electrolyte contains anionic immobile sulfonic acid groups and protons (hydronium ions) as counterions. At RH > 30%, water loading of PSS chains permits hopping of the protons between sulfonic acid groups,[10] which may lead to increased charge mobility. At the same time, water-induced swelling of the PSS matrix increases separation between adjacent PEDOT domains, reducing charge carrier concentration.[6,21] The mechanism of charge transport and how adsorption of water affects dielectric response of PEDOT:PSS film across a broad frequency range is still unclear. To examine these questions, we studied electrical and optical RH responses of PEDOT:PSS films in situ. For the first time, we complement gravimetric and optoelectronic measurements of water adsorption on the macro scale with nanoscale probing of the PEDOT:PSS surface potential to unfold connections between the rate of water adsorption and changes in electrical and optical properties of the film. Because the current study focuses on high frequency dielectric response of the PEDOT:PSS film, which is dominated by electronic conductivity, the effect of water adsorption on ionic conductivity in PEDOT:PSS films was not investigated in detail here. We will address this important question in a follow-up manuscript.

■ EXPERIMENTAL SECTION

Conductive grade PEDOT:PSS (0.5% PEDOT, 0.8% PSS by wt.) was acquired from Sigma-Aldrich. This particular composition was chosen as the subject of study because of its high conductivity and optical transparency, which makes it a material of choice for transparent electrode applications in thin film photovoltaic and top gate field-effect devices. AT-cut gold-plated 5 MHz quartz crystal microbalance (QCM) crystals and Si/SiO_2 substrates with and without silver interdigitated electrodes (IDEs) were cleaned by sonication in acetone for 30 min. PEDOT:PSS solution was spin-cast at 3000 rpm for 60 s to obtain ~50 nm thick films on all substrates. Atomic force microscope (AFM) and Kelvin probe force microscope (KPFM) images were acquired using an Asylum Research Cypher S AFM in tapping mode with conductive Pt/Ir-coated cantilever operated at 73 MHz resonance frequency. All AFM and KPFM measurements were performed inside a controlled gas environment chamber. Humidity was controlled using an L&C RH200 humidity generator by mixing water vapor with dry N_2. The film was allowed to equilibrate for 2 h before each image was acquired. Spectroscopic ellipsometry (SPE) was performed using a J. A. Woollam Co. variable-angle ellipsometer in the 450−1000 nm wavelength range. During SPE measurements, RH was controlled using a specially designed gas flow cell. A Zahner IM6 electrochemical workstation was used to perform in situ electrochemical impedance spectroscopy (EIS) during the SPE measurements. After relative humidity was increased, the film was allowed to stabilize and equilibrate with the environment for 30 min prior to DC electrical resistance, QCM, EIS, and SPE measurements.[27]

Prior to vacuum-based measurements, films were placed under vacuum of 10^{-6} Torr (1.33×10^{-6} mbar) for 3 h. Samples were exposed to increasing H_2O vapor pressures; frequency shift of the

DOI: 10.1021/acsami.7b03128
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

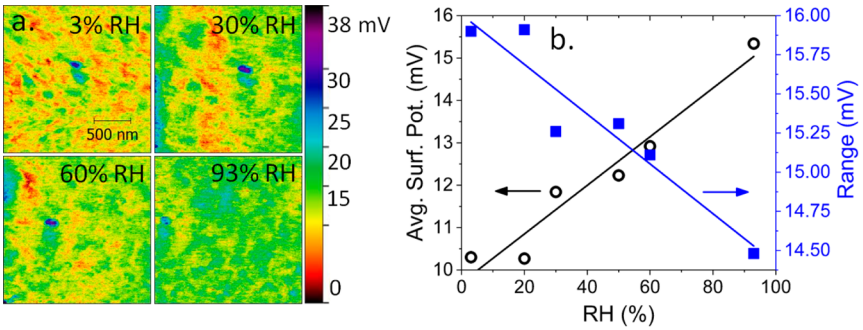


Figure 2. (a) Surface potential of PEDOT:PSS film at 3, 30, 60, and 93% RH. The image area is the same as that shown in Figure 1b inset. (b) Statistical analysis of KPFM images shown in panel a. Average surface potential increases and the range of measured surface potential values decreases as water sorption leads to homogenization of the surface.

QCM crystal was measured using an SRS200 QCM controller, and resistance measurements were performed with a Keithley 2420 sourcemeter in situ. The QCM frequency shift was corrected by subtracting the shift from a bare gold-coated QCM crystal to remove the effect of H_2O adsorption on gold. Water vapor was injected into the vacuum chamber using a mass flow controller (MFC). The MFC, sourcemeter, and QCM oscillator were controlled using LabVIEW software. Vacuum-based experiments were performed in a modified dynamic vapor sorption system from Surface Measurement Systems.

RESULTS AND DISCUSSION

An AFM image of the PEDOT:PSS film at ambient humidity is shown in Figure 1a. The film contains regions of aggregated grains resulting in a rough (root-mean-square roughness ~8.5 nm) surface. Figure 1b shows cross sections of the surface profile at 30, 50, and 93% RH. The cross-section region is represented by the white bar on the AFM image in the inset. As RH increases, the profiles show an increase in film height relative to the height of the tallest feature. The water-induced film swelling is consistent with swelling reported in other studies.[10,23] The most notable swelling occurs at RH = 93%, which is likely due to H_2O multilayer formation observed by others in PEDOT:PSS films at RH > 80%.[2,6] Because filling of small (<50 nm) interdomain regions by water is not easily resolved in the AFM image, homogenization of the film surface is best represented using a statistical approach, in which surface area, skew, and kurtosis of the images were calculated and are plotted in Figure 1c. The curved lines are exponential fits. With increasing RH, the surface area, kurtosis, and skewness in height distribution (inset) decrease. As water crowding occurs on the film surface, capillary action facilitates the filling of intergrain crevices by H_2O. The formation of an H_2O meniscus on the PEDOT:PSS surface leads to further homogenization of the surface, resulting in reduced surface area. The third and fourth standardized moments of the height distribution are skewness and kurtosis, respectively. For a Gaussian distribution, kurtosis is equal to 3σ,[4] where σ is the standard deviation. The decrease in kurtosis with increasing RH corresponds to a general homogenization of the film surface, where the height deviations of both high outliers (peaks) and low outliers (micropores) are decreased as H_2O adsorption takes place on the film. Skewness represents a measure of the asymmetry in the height distribution. Skewness >0 indicates that the histogram of height values visible in the AFM image is skewed to the right. This corresponds to a film in which the height of the primary outlier features is greater than the average film height, as opposed to a porous or pitted film surface (skewness <0). As RH increases, the skewness decreases toward 0. This indicates that film swelling reduces the difference between the average film height and the height of the tallest features, consistent with the cross-section profiles in Figure 1b.

Simultaneous gravimetric and electrical resistance measurements show that water sorption on PEDOT:PSS increases mass and electrical resistance of the film (Figure 1d). Average rate of change in mass ($\Delta M/\Delta$RH, [ng/%RH]) and rate of change in resistance ($\Delta R/\Delta$RH, [mΩ/%RH]) are shown in the inset. The values of $\Delta M/\Delta$RH and $\Delta R/\Delta$RH were calculated using the difference between adjacent points in Figure 1d to give a rough estimate of the humidity regimes in which mass adsorption and electrical resistance exhibit the highest response to changes in RH. At least 3 distinct regimes are visible between 3 and 93% RH. Both $\Delta M/\Delta$RH and $\Delta R/\Delta$RH are highest for RH < 20%. Exposure of PEDOT:PSS to H_2O vapor at low RH levels results in rapid adsorption of water at hydrophilic HSO_3 sites, leading to correlated significant change in both ΔM and ΔR as the PSS matrix undergoes initial water uptake and swelling. For 20 < RH < 60%, both $\Delta M/\Delta$RH and $\Delta R/\Delta$RH remain low, indicating that the mass loading rate slows as adsorption sites become crowded by H_2O molecules. At RH > 60%, the magnitudes of $\Delta M/\Delta$RH and $\Delta R/\Delta$RH increase by nearly a factor of 2. It is likely that this is a result of multilayer H_2O formation on the film surface after active sites become saturated with H_2O. Surface water forms hydrogen bonds with molecules in the gas phase, leading to an increase in the mass loading rate, as reported by QCM. It is likely that the decreases in $\Delta M/\Delta$RH and $\Delta R/\Delta$RH observed at RH > 80% occur as a result of equilibrium between adsorption and desorption of water molecules from the H_2O-saturated film surface.

Results of KPFM measurements are shown in Figure 2 with topography of the imaged area shown in the inset of Figure 1b. Statistical analysis of KPFM images in Figure 2b shows that as RH increases from 3 to 93%, average surface potential increases by ~50%. The charged AFM tip induces image charges in the PEDOT:PSS film which alter electrostatic potential outside the film by a term $-e/4\varepsilon x$,[28] where ε is the dielectric constant of the film and x is the distance to the film surface. Uptake of water into the PEDOT:PSS matrix increases the film's average dielectric constant (ε), which results in higher potential on the surface due to the smaller induced image charge. Surface potential variation across the measured area decreases by ~10% as RH increases from 3 to 93% (right y-axis in Figure 2b). The dipole moment of water compensates dipoles on the PEDOT:PSS surface, which decreases work function of the film and leads to screening of the polymer complex from the AFM tip, resulting in an increasingly uniform surface potential.

DOI: 10.1021/acsami.7b03128
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

Surface potential homogenization caused by water sorption was observed in oxide films as well.[29]

Water interaction with PEDOT:PSS, which leads to formation of hydronium ions, changes the concentration of mobile ions in the film. Conductivity of PEDOT:PSS is commonly explained in terms of electron/hole hopping, but it is likely that mixed ionic and electronic charge transport occurs.[10,26] Figure 3 shows the dielectric response of PEDOT:PSS film to changing RH levels in (a) Nyquist (imaginary impedance (Z'') vs real impedance(Z')) and (b) Bode plot (impedance vs frequency) representations. Symbols represent experimental data, and solid lines show fits to the equivalent circuit model. We used a modified Randles equivalent circuit for modeling multistep adsorption of water on polymer film containing both ionic and electronic charge carriers (inset in panel b).[30−32] The equivalent circuit contains contributions from shunt resistance (R_s), film capacitance (C_f), ionic charge resistance (R_{ion}), and an RC element representing diffusion/transport across boundaries with charge transfer resistance (R_{ct}) and film/electrode interface capacitance (C_e). Low frequency (<10^4 Hz) impedance increases by nearly 100% as RH is increased from 9 to 80%, likely due to swelling of the PSS matrix which increases the distance between conductive PEDOT domains.[22] A three-dimensional (3D) representation of the Nyquist plot under changing RH is shown in Figure S1, and phase shift of the film is shown in Figure S2.

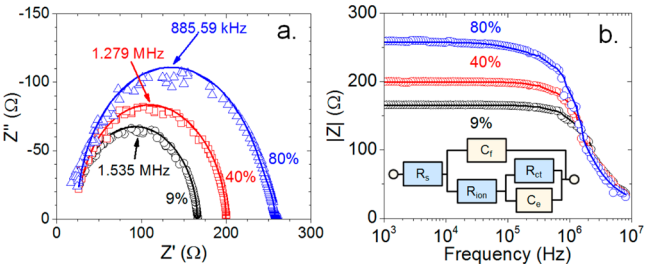


Figure 3. Nyquist (a) and Bode (b) representations of impedance spectra at 9, 40, and 80% RH. Inset in panel b shows equivalent circuit model used to fit the EIS data, where R_s = shunt resistance, C_f = film capacitance, R_{ion} = ion conduction resistance, R_{ct} = charge transfer resistance, and C_e = film/electrode interface capacitance.

Figures 4a and b show RH-dependence of real (Z') and imaginary (Z'') impedance of the PEDOT:PSS film as measured at 1 kHz, 500 kHz, and 1 MHz. From 3 to 85% RH, Z' nearly doubles at 1 kHz while staying relatively constant ($Z' \approx 120\ \Omega$) at 1 MHz. The RH dependence of Z' decreases as the frequency is increased, suggesting that the resistive RH response of PEDOT:PSS occurs primarily in the low frequency direct-current (DC) regime, which is consistent with ΔR reported in Figure 1d. The value of Z'' shows weak ($|\Delta Z''| < 6\ \Omega$) RH dependence at 1 kHz but exhibits strong RH response at higher frequencies, decreasing by over 100% at 500 kHz and by 70% at 1 MHz as RH is increased from 3 to 85%. The RH dependence of Z'' at high frequencies suggests that the change in Z'' is a consequence of the changing dielectric constant of the PEDOT:PSS film caused by inclusion of polarizable water molecules.

The derivatives of real impedance (dZ') and imaginary impedance (dZ'') at 1 kHz, 500 kHz, and 1 MHz are shown in Figure S3. In general, the dZ' and dZ'' responses can be divided into at least three regions which resemble the regimes identified for string changes in ΔM and ΔR shown in Figure 1d: RH < 20%, 20% < RH < 60%, and RH > 60%. In most cases, dZ' and dZ'' change by 100% or more for 3% < RH < 20%, which is likely associated with rapid adsorption of H_2O at hydrophilic HSO_3 surface sites. As RH approaches 20%, dZ' and dZ'' become relatively constant as low adsorption energy sites become saturated with water. Near 60% RH, a dramatic change in dZ' and dZ'' occurs, which likely corresponds to the same process responsible for changes in ΔM and ΔR at this RH level: saturation of the bulk film and formation of H_2O multilayers on the film surface.

RH dependence of equivalent circuit parameters is shown in Figures 4c and d. Shunt resistance increases insignificantly (from 23 to 29 Ω) within the full measured RH range. Charge

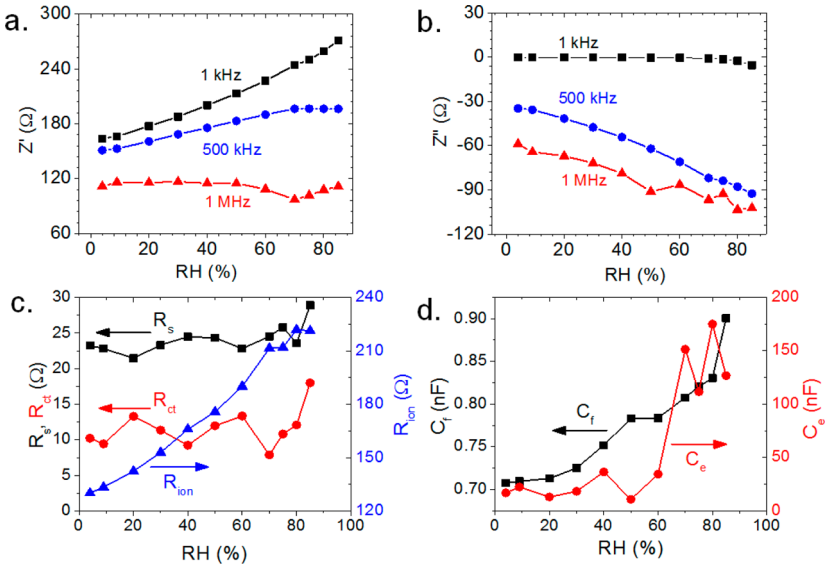


Figure 4. (a) Frequency-dependent real (Z') and (b) imaginary (Z'') impedance measured at 1 kHz, 500 kHz, and 1 MHz. (c) Equivalent circuit shunt resistance (R_s), ionic conduction resistance (R_{ion}), and charge transfer resistance (R_{ct}), and (d) film capacitance (C_f) and film/electrode interface capacitance (C_e).

DOI: 10.1021/acsami.7b03128
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

transfer resistance is not affected until RH > 75%, likely due to condensation of water between domains at high RH. From 3 to 85% RH, ion conduction resistance increases by ∼70% (from 130 to 220 Ω). Evidence by Nardes et al. that ionic conduction channels begin contributing to conductivity of PEDOT:PSS at RH > 30% due to proton hopping between sulfonic acid groups[10] suggests that the increase in ionic conduction resistance that we observe is due to a decrease in mobile ion concentration due to film swelling and formation of hydration spheres around mobile ions. We find that the interface capacitance and film capacitance increase significantly with RH. Capacitance at film/electrode interfaces exhibits a large increase at RH > 60%, likely as a result of water condensation, which allows formation of metastable electrical double layer (EDL) structures comprised of hydronium ions (H_3O^+) and free hydrogen ions (H^+) generated by H_2O adsorption on hydrophilic PSS:H_2O + PSS(HSO_3) → H_3O^+PSS$(SO_3)^-$[31]. Film capacitance increases from 0.71 to 0.90 nF, which is closely related to the high dielectric constant of H_2O (∼80) relative to the dielectric constant of PEDOT:PSS (∼2−3).[33] It is likely that some of the observed increase in ion conduction resistance results from electric field screening of ions by polar water molecules.

We explored changes in the optical properties of PEDOT:PSS film deposited on SiO_2 using variable-angle SPE under controlled RH conditions. SPE spectra measured at different RH levels were modeled in 450−1000 nm wavelength region to omit higher energy transitions of the spectrum, which include Si/SiO_2 contributions. A single layer of Tauc−Lorentz + Lorentz oscillators was used to model the PEDOT:PSS film. We evaluated the fit parameters for sensitivity (confidence limits) and correlations (two parameter correlation coefficients). The errors reported on the parameters were derived from $FOM_i = 1.65\sqrt{C_{ii}}\sqrt{MSE}$ where MSE is the mean squared error and C_{ii} is a two parameter correlation coefficient. The rates of change in direct and indirect optical bandgaps ($\Delta E_g/\Delta RH$) of the PEDOT:PSS film, as extracted from the SPE spectra, are shown in Figure 5 (see Figure S4 for optical bandgap calculation). The values of $\Delta E_g/\Delta RH$ exhibit at least 3 different trends under changing RH conditions: the magnitude of $\Delta E_g/\Delta RH$ decreases by over 60% for 1 < RH < 10%, remains roughly constant for 10 < RH < 75%, and increases for RH > 75%. These 3 regimes correspond to similar transitions observed in the ΔM, ΔR, and EIS data, which generally show a rapid change at low RH (1 < RH < 10−25%), stabilized change at intermediate RH (10−25 < RH < 60−75%), and increased change at high RH (RH > 60−75%). This suggests that the same physical PEDOT:PSS-H_2O interactions are responsible for changes in rates of mass loading, conductivity, and optical (bandgap) responses. It is likely that the three regimes roughly correspond to rapid adsorption of H_2O at active sites at low RH, the slow adsorption and diffusion into the bulk film at mid-RH, and H_2O saturation and multilayer formation on the surface at high RH. Optical film thickness increases exponentially with RH (Figure 5 inset). Exponential swelling of the film is consistent with adsorption that resembles a type-III BET isotherm,[34] suggesting that hydrogen bonding between gas-phase H_2O and adsorbed H_2O molecules leads to an increase in the rate of adsorption and film swelling as RH increases, which is consistent with previous observations of H_2O interaction with PEDOT:PSS film[15]

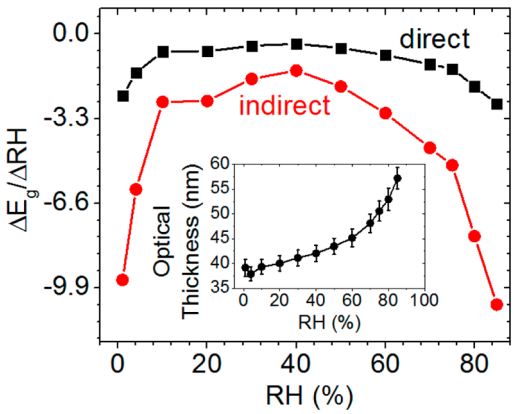


Figure 5. Rate of change in the optical bandgap (E_g) as estimated from spectroscopic ellipsometry. Units of $\Delta E_g/\Delta RH$ are in eV/%. Inset: optical thickness of PEDOT:PSS film as estimated from fit of SPE spectra. Error bars represent confidence limits estimated from MSE values of the fits.

Wavelength-dependent real (ε_1) and imaginary (ε_2) dielectric constants, index of refraction (n), and extinction coefficient (k) of PEDOT:PSS at 1 and 85% RH are shown in Figure S5. For λ > 500 nm, real dielectric constant/index of refraction increase with RH as H_2O loading of the PEDOT:PSS matrix increases polarizability of the film. RH dependence of the real dielectric constant/index of refraction is insignificant at λ < 500 nm due to low dispersion and optical extinction coefficient (<0.05). The imaginary dielectric constant/extinction coefficient exhibit a bathochromic shift of ∼75 nm as RH increases from 1 to 85%, indicating the creation of lower energy states. The peak shift corresponds to $\Delta E \approx 16$ eV, which is likely related to the formation of EDLs comprised of H_3O^+ and H^+ ions between adjacent PEDOT:PSS domains. Formation of EDLs increases interdomain capacitance and inhibits ion mobility,[35] which is apparent from the large increases in R_{ion} and C_f shown in Figure 4. It is likely that the formation of metastable EDLs also contributes to film swelling (Figure 5 inset) and the increased PEDOT:PSS surface potential at high RH (Figure 2).

To examine connections between electronic and optical responses of the PEDOT:PSS film under changing humidity, we extracted the real part of optical conductivity (σ_{op}) from the imaginary dielectric constant (ε_2) and optical frequency (ω), where $\sigma_{op} = \varepsilon_2\omega/4\pi$. Wavelength-dependent optical conductivity of the film at different RH conditions is shown in Figure 6a. As RH increases, optical conductivity decreases nonlinearly. The decrease in σ_{op} at λ > 700 nm is likely due to the high optical extinction coefficient of water at near-infrared wavelengths. Because electrical conductivity is proportional to $1/R$, where R is the DC electrical resistance shown in Figure 1d, it is possible to examine the RH dependence of optical and electrical conductivities together. To compare the PEDOT:PSS optical conductivity response to RH with electrical conductivity response to RH, we plotted σ_{op} as measured at λ = 500 nm with $1/R$ (Figure 6b). The values of σ_{op} and $1/R$ appear strongly correlated and exhibit behavior similar to that observed for the rate of change of optical band gap ($\Delta E_g/\Delta RH$) and the rate of mass loading ($\Delta M/\Delta RH$). Both electrical and optical conductivities decrease roughly exponentially as humidity is increased. Mobility of electronic charges decreases when swelling of the hydrophilic PSS matrix results in larger distances between adjacent conductive PEDOT domains. We

DOI: 10.1021/acsami.7b03128
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

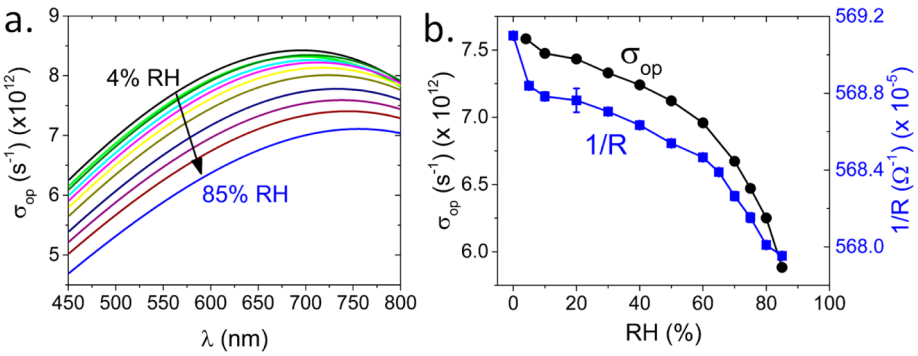


Figure 6. (a) Wavelength-dependent optical conductivity (σ_{op}) extracted from imaginary dielectric constant measured at different RH conditions. (b) Strong correlation between optical conductivity (extracted at 550 nm) and electrical conductivity, represented by inverse of DC resistance (1/*R*).

speculate that film swelling along with the formation of EDL structures composed of H_3O^+ and H^+ ions results in reduced carrier mobility, which concomitantly lowers electrical and optical conductivities in the film.

CONCLUSIONS

We performed multimodal probing of the PEDOT:PSS-H_2O interaction using nanoscale and mesoscale characterization to unfold connections between the electronic, ionic, optical, and structural humidity response of PEDOT:PSS films. We showed a direct correlation between decreases in electrical and optical conductivities as RH increases. Mass loading, DC electrical conductivity, optical conductivity, and optical bandgap changes in the film occurred in three primary RH regimes. Large changes occurred at low (RH < 10−25%) and high (RH > 60−75%) humidity conditions, and small changes occurred at mid-RH conditions (25 < RH < 60%). The three regimes likely correspond to physical processes taking place in the film: rapid H_2O adsorption at active PSS sites at low RH, H_2O adsorption and diffusion throughout the bulk film at mid-RH, and H_2O saturation and multilayer formation on the film surface at high RH. As calculated from SPE spectra, optical film thickness swelled to 150% of its original thickness from 1 to 85% RH. Low frequency (1 kHz) impedance increased by ∼100% and film capacitance increased by ∼30% as RH was increased from 9 to 80%. Our results indicate that this was likely caused by a combination of film swelling, an increase in the average dielectric constant due to uptake of water, and formation of electrical double layers by mobile hydronium ions and protons. The results shed light on the correlation between nanoscale surface properties and mesoscale optoelectronic behavior of PEDOT:PSS under varying humidity conditions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.7b03128.

Impedance spectroscopy data and information related to optical bandgap calculation and dielectric constants (PDF)

AUTHOR INFORMATION

Corresponding Authors

*E-mail: muckleyes@ornl.gov.
*E-mail: ivanovin@ornl.gov.

ORCID

Eric S. Muckley: 0000-0001-7114-5424
Rajeev Kumar: 0000-0001-9494-3488
Bobby G. Sumpter: 0000-0001-6341-0355

Funding

[This manuscript was authored by UT-Battelle, LLC under Contract DE-AC05-00OR22725 with the U.S. Department of Energy. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for United States Government purposes. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (http://energy.gov/downloads/doe-public-access-plan).

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research was conducted at the Center for Nanophase Materials Sciences, which is a DOE Office of Science User Facility. K.V. acknowledges support from the Science Undergraduate Laboratory Internships (SULI) program of the DOE. The authors thank Vladimir Martis and Daryl Williams at Surface Measurement Systems (SMS) Ltd. for their support in development of new capabilities of multimodal environmental system from SMS.

REFERENCES

(1) Tang, S. L. P. Recent Developments in Flexible Wearable Electronics for Monitoring Applications. *Trans. Inst. Meas. Control* **2007**, *29* (3−4), 283−300.

(2) Kuş, M.; Okur, S. Electrical Characterization of Pedot: Pss Beyond Humidity Saturation. *Sens. Actuators, B* **2009**, *143* (1), 177−181.

(3) Eom, S. H.; Senthilarasu, S.; Uthirakumar, P.; Yoon, S. C.; Lim, J.; Lee, C.; Lim, H. S.; Lee, J.; Lee, S.-H. Polymer Solar Cells Based on Inkjet-Printed Pedot: Pss Layer. *Org. Electron.* **2009**, *10* (3), 536−542.

(4) Vosgueritchian, M.; Lipomi, D. J.; Bao, Z. Highly Conductive and Transparent Pedot: Pss Films with a Fluorosurfactant for Stretchable and Flexible Transparent Electrodes. *Adv. Funct. Mater.* **2012**, *22* (2), 421−428.

(5) Taccola, S.; Greco, F.; Zucca, A.; Innocenti, C.; de Julián Fernández, C. s.; Campo, G.; Sangregorio, C.; Mazzolai, B.; Mattoli, V. Characterization of Free-Standing Pedot: Pss/Iron Oxide Nanoparticle Composite Thin Films and Application as Conformable Humidity Sensors. *ACS Appl. Mater. Interfaces* **2013**, *5* (13), 6324−6332.

(6) Huang, J.; Miller, P. F.; Wilson, J. S.; de Mello, A. J.; de Mello, J. C.; Bradley, D. D. Investigation of the Effects of Doping and Post -Deposition Treatments on the Conductivity, Morphology, and Work Function of Poly (3, 4-Ethylenedioxythiophene)/Poly (Styrene Sulfonate) Films. *Adv. Funct. Mater.* **2005**, *15* (2), 290−296.
(7) Kawano, K.; Pacios, R.; Poplavskyy, D.; Nelson, J.; Bradley, D. D.; Durrant, J. R. Degradation of Organic Solar Cells Due to Air Exposure. *Sol. Energy Mater. Sol. Cells* **2006**, *90* (20), 3520−3530.
(8) Yun, J. M.; Yeo, J. S.; Kim, J.; Jeong, H. G.; Kim, D. Y.; Noh, Y. J.; Kim, S. S.; Ku, B. C.; Na, S. I. Solution-Processable Reduced Graphene Oxide as a Novel Alternative to Pedot: Pss Hole Transport Layers for Highly Efficient and Stable Polymer Solar Cells. *Adv. Mater.* **2011**, *23* (42), 4923−4928.
(9) Sun, L.; Wang, J.; Butt, H. J.; Bonaccurso, E. Influence of Relative Humidity on the Nanoscopic Topography and Dielectric Constant of Thin Films of Ppy: Pss. *Small* **2011**, *7* (7), 950−956.
(10) Nardes, A. M.; Kemerink, M.; De Kok, M.; Vinken, E.; Maturova, K.; Janssen, R. Conductivity, Work Function, and Environmental Stability of Pedot: Pss Thin Films Treated with Sorbitol. *Org. Electron.* **2008**, *9* (5), 727−734.
(11) Bailo, D.; Generosi, A.; Albertini, V. R.; Caminiti, R.; De Bettignies, R.; Paci, B. Time-Resolved Morphological Study of 'Pedot: Pss'hole Transporting Layer for Polymer Solar Cells. *Synth. Met.* **2012**, *162* (9), 808−812.
(12) Wang, J.; Zheng, Z.; Li, H.; Huck, W.; Sirringhaus, H. Dewetting of Conducting Polymer Inkjet Droplets on Patterned Surfaces. *Nat. Mater.* **2004**, *3* (3), 171−176.
(13) Lang, U.; Naujoks, N.; Dual, J. Mechanical Characterization of Pedot: Pss Thin Films. *Synth. Met.* **2009**, *159* (5), 473−479.
(14) Yang, Y.; Lee, K.; Mielczarek, K.; Hu, W.; Zakhidov, A. Nanoimprint of Dehydrated Pedot: Pss for Organic Photovoltaics. *Nanotechnology* **2011**, *22* (48), 485301.
(15) Muckley, E. S.; Lynch, J.; Kumar, R.; Sumpter, B.; Ivanov, I. N. Pedot: Pss/Qcm-Based Multimodal Humidity and Pressure Sensor. *Sens. Actuators, B* **2016**, *236*, 91.
(16) Jiang, Y.; Luo, B.; Jiang, F.; Jiang, F.; Fuentes-Hernandez, C.; Liu, T.; Mao, L.; Xiong, S.; Li, Z.; Wang, T. Efficient Colorful Perovskite Solar Cells Using a Top Polymer Electrode Simultaneously as Spectrally Selective Antireflection Coating. *Nano Lett.* **2016**, *16* (12), 7829−7835.
(17) Tong, J.; Xiong, S.; Zhou, Y.; Mao, L.; Min, X.; Li, Z.; Jiang, F.; Meng, W.; Qin, F.; Liu, T. Flexible All-Solution-Processed All-Plastic Multijunction Solar Cells for Powering Electronic Devices. *Mater. Horiz.* **2016**, *3* (5), 452−459.
(18) Okuzaki, H.; Kuwabara, T.; Funasaka, K.; Saido, T. Humidity-Sensitive Polypyrrole Films for Electro-Active Polymer Actuators. *Adv. Funct. Mater.* **2013**, *23* (36), 4400−4407.
(19) Taccola, S.; Greco, F.; Sinibaldi, E.; Mondini, A.; Mazzolai, B.; Mattoli, V. Toward a New Generation of Electrically Controllable Hygromorphic Soft Actuators. *Adv. Mater.* **2015**, *27* (10), 1668−1675.
(20) Elschner, A.; Kirchmeyer, S.; Lövenich, W.; Merker, U.; Reuter, K. *Pedot Principles and Applications of an Intrinsically Conductive Polymer*, 1st ed.; Taylor and Francis Group: Boca Raton, 2011.
(21) Stöcker, T.; Köhler, A.; Moos, R. Why Does the Electrical Conductivity in Pedot: Pss Decrease with Pss Content? A Study Combining Thermoelectric Measurements with Impedance Spectroscopy. *J. Polym. Sci., Part B: Polym. Phys.* **2012**, *50* (14), 976−983.
(22) Dupont, S. R.; Novoa, F.; Voroshazi, E.; Dauskardt, R. H. Decohesion Kinetics of Pedot: Pss Conducting Polymer Films. *Adv. Funct. Mater.* **2014**, *24* (9), 1325−1332.
(23) Friedel, B.; Keivanidis, P. E.; Brenner, T. J.; Abrusci, A.; McNeill, C. R.; Friend, R. H.; Greenham, N. C. Effects of Layer Thickness and Annealing of Pedot: Pss Layers in Organic Photodetectors. *Macromolecules* **2009**, *42* (17), 6741−6747.
(24) Okuzaki, H.; Suzuki, H.; Ito, T. Electrically Driven Pedot/Pss Actuators. *Synth. Met.* **2009**, *159* (21), 2233−2236.
(25) Okuzaki, H.; Hosaka, K.; Suzuki, H.; Ito, T. SPIE Smart Structures and Materials+ Nondestructive Evaluation and Health Monitoring, International Society for Optics and Photonics. *Proc. SPIE* **2010**, 76422U−76422U-8.
(26) Rivnay, J.; Inal, S.; Collins, B. A.; Sessolo, M.; Stavrinidou, E.; Strakosas, X.; Tassone, C.; Delongchamp, D. M.; Malliaras, G. G. Structural Control of Mixed Ionic and Electronic Transport in Conducting Polymers. *Nat. Commun.* **2016**, *7*, 11287.
(27) Muckley, E. S.; Lynch, J.; Kumar, R.; Sumpter, B.; Ivanov, I. N. Pedot: Pss/Qcm-Based Multimodal Humidity and Pressure Sensor. *Sens. Actuators, B* **2016**, *236*, 91−98.
(28) Emtage, P.; O'Dwyer, J. Richardson-Schottky Effect in Insulators. *Phys. Rev. Lett.* **1966**, *16* (9), 356.
(29) Jacobs, C. B.; Ievlev, A. V.; Collins, L. F.; Muckley, E. S.; Joshi, P. C.; Ivanov, I. N. SPIE OPTO, International Society for Optics and Photonics. *Proc. SPIE* **2016**, 97491Q−97491Q-6.
(30) Mostany, J.; Scharifker, B. Impedance Spectroscopy of Undoped, Doped and Overoxidized Polypyrrole Films. *Synth. Met.* **1997**, *87* (3), 179−185.
(31) Harrington, D. A. The Rate-Determining Step in Electrochemical Impedance Spectroscopy. *J. Electroanal. Chem.* **2015**, *737*, 30−36.
(32) Doan, T. C.; Ramaneti, R.; Baggerman, J.; Tong, H. D.; Marcelis, A. T.; van Rijn, C. J. Intrinsic and Ionic Conduction in Humidity-Sensitive Sulfonated Polyaniline. *Electrochim. Acta* **2014**, *127*, 106−114.
(33) Yamashita, M.; Otani, C.; Shimizu, M.; Okuzaki, H. Effect of Solvent on Carrier Transport in Poly (3, 4-Ethylenedioxythiophene)/Poly (4-Styrenesulfonate) Studied by Terahertz and Infrared-Ultraviolet Spectroscopy. *Appl. Phys. Lett.* **2011**, *99* (14), 143307.
(34) Brunauer, S.; Deming, L. S.; Deming, W. E.; Teller, E. On a Theory of the Van Der Waals Adsorption of Gases. *J. Am. Chem. Soc.* **1940**, *62* (7), 1723−1732.
(35) Tai, Y.; Lubineau, G. Heating-Rate-Triggered Carbon-Nanotube-Based 3-Dimensional Conducting Networks for a Highly Sensitive Noncontact Sensing Device. *Sci. Rep.* **2016**, *6*, 1 DOI: 10.1038/srep19632.

DOI: 10.1021/acsami.7b03128
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX